

Global and Local Gauge Transformations

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Resumen

These notes explain the mathematical distinction between global and local gauge transformations. The gauge group G is kept fixed, while the difference lies in whether the transformation is represented by a single group element or by a smooth spacetime-dependent map into the group. The case $G = U(1)$ is used to show why local gauge symmetry requires the introduction of a gauge field and a covariant derivative.

Global and Local Gauge Transformations

A gauge theory is characterized by a Lie group G , called the *gauge group*. The distinction between a *global* and a *local* symmetry does not lie in the choice of G , but rather in how the elements of G are allowed to act on the fields.

Global Symmetry

A global transformation is specified by a single, spacetime-independent group element,

$$g \in G, \quad g = \text{constant}. \quad (1)$$

If a matter field $\phi(x)$ transforms according to a representation of G , then

$$\boxed{\phi(x) \mapsto g \phi(x)}. \quad (2)$$

For the group $U(1)$, the group element can be written as

$$g = e^{i\alpha}, \quad \alpha = \text{constant}, \quad (3)$$

so that

$$\boxed{\phi(x) \mapsto e^{i\alpha} \phi(x)}. \quad (4)$$

Since the transformation parameter is constant,

$$\partial_\mu \alpha = 0,$$

the ordinary derivative transforms covariantly,

$$\partial_\mu \phi \mapsto g \partial_\mu \phi, \quad (5)$$

and no additional fields are required.

Local Gauge Symmetry

A local gauge transformation allows a different group element at every point of spacetime. Mathematically, the transformation is a smooth map

$$\boxed{g : M \longrightarrow G,} \quad (6)$$

where M denotes spacetime.

The field transforms according to

$$\boxed{\phi(x) \longmapsto g(x)\phi(x).} \quad (7)$$

For the case $G = U(1)$,

$$g(x) = e^{i\alpha(x)}, \quad (8)$$

and therefore

$$\boxed{\phi(x) \longmapsto e^{i\alpha(x)}\phi(x).} \quad (9)$$

Because the transformation parameter now depends on spacetime,

$$\partial_\mu \alpha(x) \neq 0,$$

the ordinary derivative transforms as

$$\partial_\mu \phi \longmapsto e^{i\alpha(x)} [\partial_\mu \phi + i(\partial_\mu \alpha)\phi], \quad (10)$$

which is no longer covariant.

To restore covariance one introduces a gauge field A_μ and defines the covariant derivative

$$\boxed{D_\mu = \partial_\mu + ieA_\mu.} \quad (11)$$

For a $U(1)$ gauge theory, the gauge field transforms as

$$\boxed{A_\mu \longmapsto A'_\mu = A_\mu - \frac{1}{e}\partial_\mu \alpha(x),} \quad (12)$$

so that

$$D_\mu \phi \longmapsto e^{i\alpha(x)} D_\mu \phi, \quad (13)$$

making the kinetic term gauge invariant.

The Case of $U(1)$

It is common to hear the phrases “global $U(1)$ ” and “local $U(1)$,” but strictly speaking $U(1)$ itself is neither global nor local. Rather, $U(1)$ is simply the symmetry group. The distinction is how its elements are chosen:

$$\text{Global } U(1) : \quad g = e^{i\alpha}, \quad \alpha = \text{constant}, \quad (14)$$

$$\text{Local } U(1) : \quad g(x) = e^{i\alpha(x)}, \quad \alpha : M \rightarrow \mathbb{R}. \quad (15)$$

Equivalently, one may say that

$$\begin{array}{ll}
\text{Global symmetry:} & G = U(1), \\
\text{Local gauge symmetry:} & \mathcal{G} = C^\infty(M, U(1)),
\end{array}
\tag{16}$$

where $C^\infty(M, U(1))$ denotes the group of smooth maps from spacetime into $U(1)$, with pointwise multiplication.

Summary

	Global	Local
Transformation	$g \in G$	$g : M \rightarrow G$
Parameter	constant	depends on x
Matter field	$\phi \mapsto g\phi$	$\phi \mapsto g(x)\phi$
Ordinary derivative	covariant	not covariant
Gauge field	not required	required

The essential mathematical distinction is therefore that a global symmetry is represented by a single element of the group G , whereas a local gauge symmetry is represented by a smooth map from spacetime into G .